Letter Frequency Attack on Additive Cipher

39. Write a C program that can perform a letter frequency attack on an additive cipher without human intervention. Your software should produce possible plaintexts in rough order of likelihood. It would be

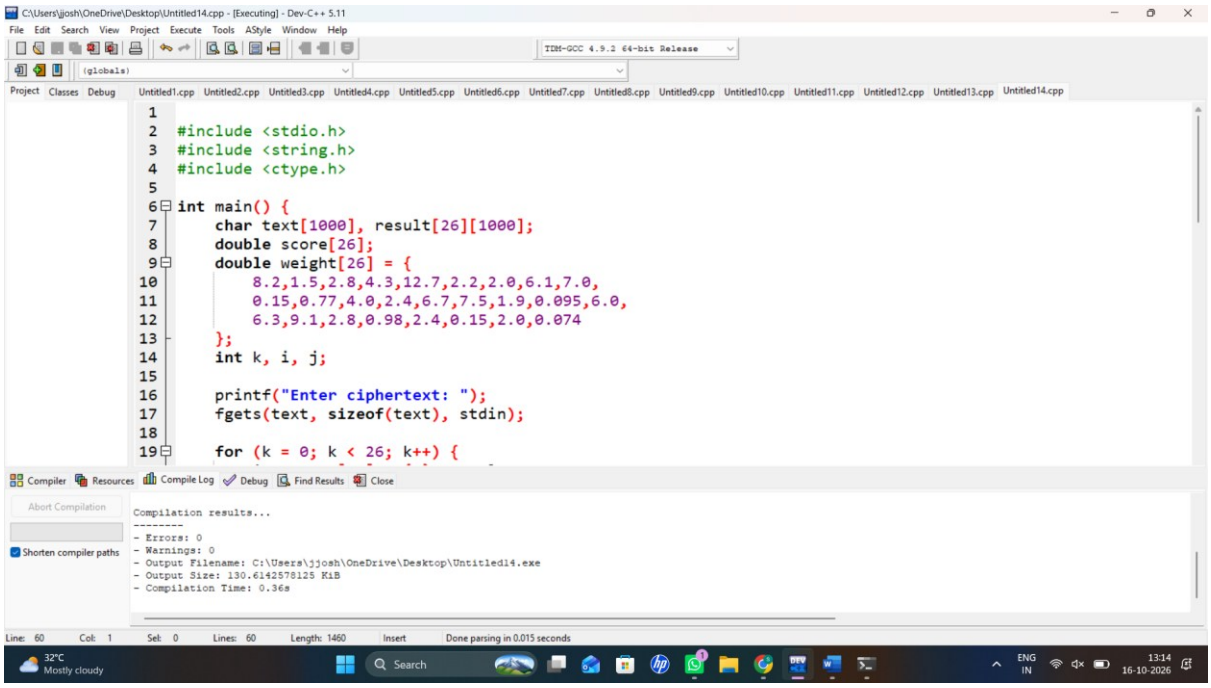
Aim:

To implement a frequency-analysis attack in C that tries all possible keys of an additive cipher and displays the top ten candidate plaintexts.

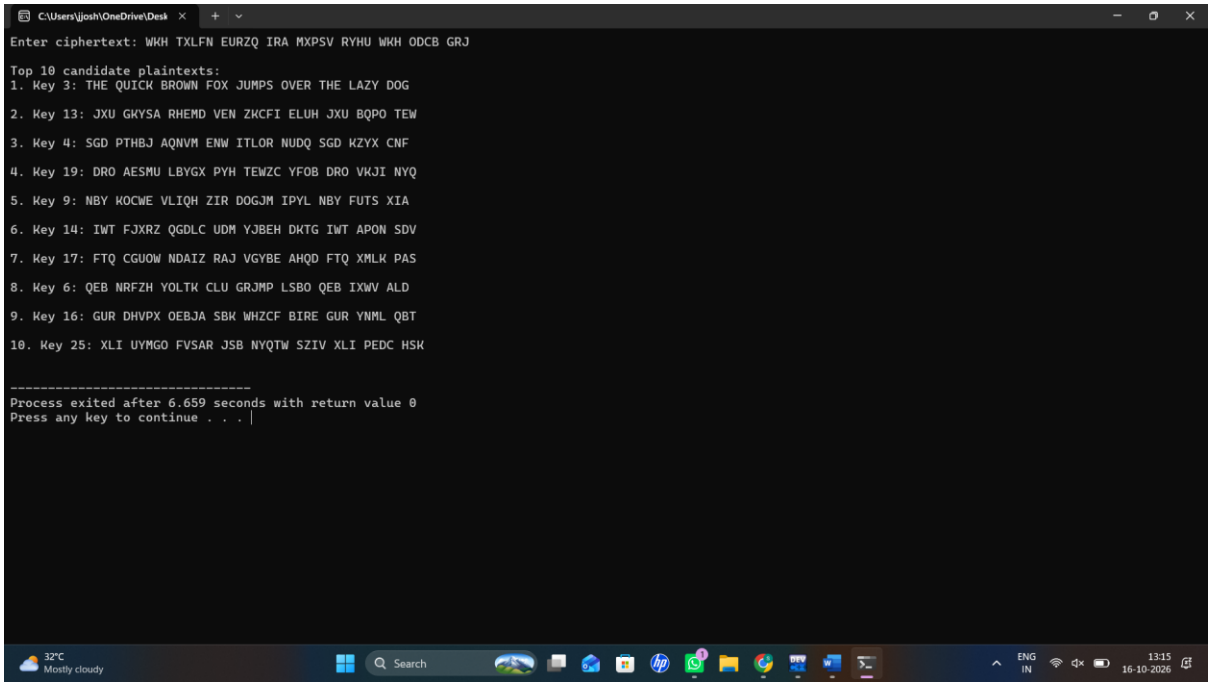
Algorithm:

1. Read the ciphertext.
2. Consider all 26 possible additive cipher keys.
3. Decrypt each character using \(P=(C-k+26)\bmod26\).
4. Calculate a score based on common English letters.
5. Sort the candidates according to their scores.
6. Display the top ten possible plaintexts.

Program:



Input/Output:



he program prints all ten candidates and their keys. The ordering is based on a simple letter-frequency score, so the actual ranking may differ from this illustrative output. A more accurate attack would score complete words and letter sequences.

Result:

The additive cipher was attacked by testing possible shifts automatically. The method demonstrates how letter-frequency analysis can help recover plaintext without knowing the encryption key.